

A results-driven **AI Engineer** with **3+ years** of hands-on experience in **Deep Learning** and **Machine Learning** (2023 — present), including **1.5 years** of professional experience in AI for **Medical Imaging**. Currently working as **Technical Staff at TANUH (AICoE), IISc**, developing **vascular disease screening** models that detect **systemic diseases** from fundus images. I build, fine-tune, and debug **PyTorch** models for **medical image segmentation**, classification, **denoising**, and **image translation**, including foundation-model fine-tuning (**RETFound**, **MedGemma**) with **LoRA**. My interests include **systemic disease detection** and **fundus & OCT imaging**.

EDUCATION

Bachelor of Technology <i>Rajiv Gandhi University of Knowledge Technologies</i> Pre-University Course (Intermediate / MPC) <i>Rajiv Gandhi University of Knowledge Technologies</i> Secondary School Certificate (10th Class) <i>BSE AP</i>	2022 — APRIL 2026 <i>CGPA: 8.5</i> 2020 — 2022 <i>CGPA: 9.56</i> 2020
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- Secured the **State 64th Rank** in the 2020 Polytechnic Entrance Examination immediately following 10th grade.

PUBLICATIONS & PRESENTATIONS

"Energy-Regularized Self-Supervised Denoising for Structure-Preserving OCT Imaging" <i>Optics Letters Journal</i>	MARCH 2026 <i>Published</i>
Addressed the over-smoothing problem in S2SNet for the OCT dataset by introducing energy regularization and validated improvements through a comprehensive comparison of segmentation performance.	
"Data Driven Approach in Building Consumer-Friendly Business Models Using Machine Learning Techniques" <i>IEEE AICECS Conference</i>	DEC 2024 <i>Manipal Institute of Technology</i>
Presented a research paper comparing three machine learning models on real-time dataset metrics.	

SKILLS

Programming & Frameworks	Python (PyTorch, PyTorch Lightning, TensorFlow, Scikit-learn, MONAI), MATLAB, OpenCV, C/C++, HF Transformers, ComfyUI
LLMs & Generative AI	Vision Language Models (VLMs) & Large Language Models (LLMs) — basic knowledge with some hands-on experience; Diffusion Models (DDPM), VAEs, GANs, Fine-tuning
DL Architectures	Transformer-based models (ViTs), Flash Attention, UNet, Multi-task & Multi-modal Learning
Imaging Systems & Infra.	Medical Imaging (MRI, CT, Fundus, OCT), DICOM/NIfTI formats, Model Optimization, Edge/Mobile Architectures
CV & Processing	Image/Video Restoration, Enhancement, Denoising, Segmentation, Object Detection, OCR, Feature Extraction
Distributed Training & Infra.	DeepSpeed, torchrun, PyTorch, PyTorch Lightning DDP, GPU Acceleration, bfloat16, DGX Spark Clusters
Inference Optimization	Model Quantization (GGUF), Knowledge Distillation, vLLM, TensorRT
Algorithms & Concepts	Regression, Decision Trees, Random Forest, Gradient Boosting, Clustering, PCA, Neural Networks, CNNs
Experiment Tracking & DevTools	MLflow, TensorBoard, Python Logger, Git CLI
Communication	English, Telugu (Native), Hindi, Technical Documentation

TECHNICAL EXPERIENCE

Technical Staff / TANUH (AICoE) IISc <i>Vascular Disease Screening Vertical</i>	JUNE 2026 — PRESENT <i>Bangalore, KA</i>
- Working in the Vascular disease screening vertical to detect systemic diseases from only 4 fundus images per patient, leveraging RETFound, MedGemma 1.5, and Qwen 3.8 27B with LoRA fine-tuning. - Built an agent-based automated data quality check pipeline to validate incoming clinical imaging data at scale. - Developed and operated a data collection and anonymisation website (developer & IT owner) hosted on Google Cloud Platform (GCP) across India for real-time data collection from hospitals. - Administered the team's compute nodes and Open OnDemand software, managing access and workloads for researchers.	

Engineering Intern / TANUH (AICoE) IISc <i>Medical Assistant and Early Disease Detection</i>	NOV 2025 — MAY 2026 <i>Bangalore, KA</i>
- Working at TANUH (AI Centre of Excellence) under the Ministry of Education) to develop AI-driven diagnostics and screening tools for early breast cancer detection and risk prediction. - Developing "Pseudohealth" generative systems to synthesize mammogram images and finetuned MedGemma 1.5 4B to work with mammogram images, transforming clinical care through scalable AI solutions.	
Developer / Video Masking & Annotation Application <i>Full-Stack AI Deployment</i>	AUG 2025 — NOV 2025 <i>Bangalore, KA</i>
Designed a medical video annotation tool integrating the Segment Anything Model (SAM) for frame-by-frame analysis, deployed via a scalable serverless architecture on Google Cloud Run.	
Research Intern / MIGLab [Link], CDS Department at IISc <i>Medical Imaging & Deep learning</i>	MARCH 2025 — OCT 2025 <i>Bangalore, KA</i>
- Executed distributed training via torchrun across DGX Spark clusters and optimized model inference through Quantization for real-time serving. - Implemented Flash Attention and Maximum Representative Subsampling (MRS) pipelines for the Federated Tumor Segmentation (FeTS) dataset to train 3D UNet models, optimizing data quality and training performance for large-scale vision tasks. - Developed OCT image denoising algorithms and currently developing model-based Quantitative Susceptibility Mapping (QSM) reconstruction algorithms, focusing on preserving structural details and high-precision accuracy.	
Teaching Assistant / Aster Academy Cohort-5 <i>AI in Healthcare / Medical Image Analysis</i>	2025 <i>Bangalore, KA</i>
Supported instruction and technical experiments for a course on AI in Healthcare focused on medical image analysis.	
Machine Learning Intern / Academor (Flutura)	JUNE 2023 — AUG 2023 <i>Remote</i>
Gained professional experience in applied machine learning, focusing on model internals, data curation, and version-controlled experimentation for industrial use cases.	

ACADEMIC PROJECTS

Deployment / Real-time Median-nerve Segmentation	JUNE 2025 — AUG 2025
Deployed VisTr on Jetson Orin NX using NVIDIA SDK Manager, while optimizing transformer performance with Flash-Attn 2 and scaling training via PyTorch DDP on dual RTX A6000 GPUs.	
Experimentation / Stable Diffusion 3.5 Medium & ComfyUI	2024
Executed inferencing of Stable Diffusion 3.5 to generate high-quality images from text prompts utilizing the ComfyUI node-based interface on Kaggle infrastructure.	
Developer / AgriBot (Collab with ICAR, Pune)	JAN 2024 — AUG 2024
Developed an end-to-end multimodal application for leaf disease detection and crop recommendation, integrating vision capabilities with Arduino, ESP, and Raspberry Pi for real-time inference.	

ACTIVITIES

- Regularly analyze **Q1/Q2 journals** and state-of-the-art (**SOTA**) research to stay at the forefront of AI advancements.
- Participant, IEEE YESIST'12 Competition (2024); National Service Scheme (NSS), 2021 — 2024.

RELEVANT CERTIFICATIONS

Machine Learning for Engineering: NPTEL / IIT Madras [Link]	2025
Python (Basic) Certified: HackerRank [Link]	2025
Essential Mathematics for ML: NPTEL / IIT Roorkee [Link]	2024
Deep Learning Specialist (PyTorch, TensorFlow, Keras): edX / IBM [Link]	2024
Geoprocessing using Python: IIRS / ISRO [Link]	2023
Intro to Machine Learning: Kaggle [Link]	2023